

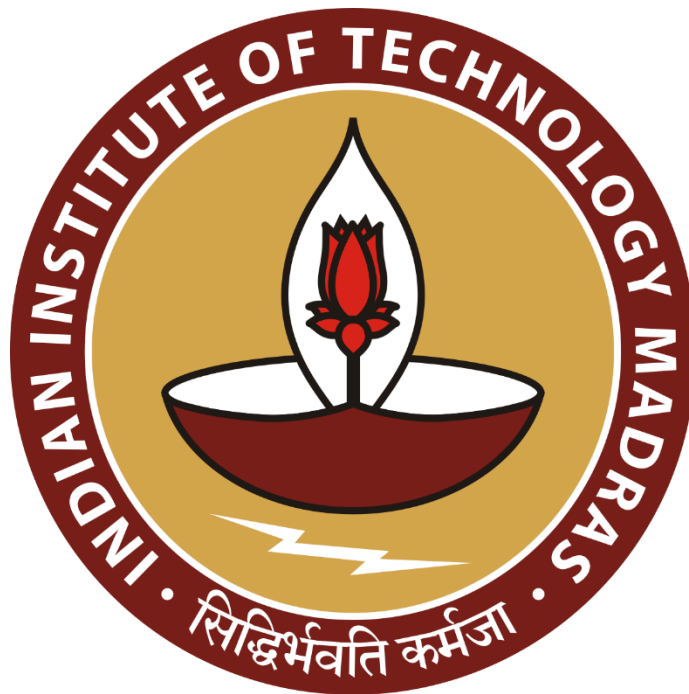
Analysing Delivery Reliability, Promotion Effectiveness and loyalty programme value for a Large E-Commerce Retailer

A Proposal report for the BDM capstone Project

Submitted by

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Contents

1 Executive Summary	3
2 Organisation Background	3
3 Problem Statement, Background and Objectives	4
4 Problem Solving Approach	5
5 Expected Timeline	6
5.1 Work Breakdown Structure	6
5.2 Gantt Chart	7
6 Expected Outcome	7

Declaration Statement

I am working on a project titled “**Analysing Delivery Reliability, Promotion Effectiveness and loyalty programme value for a Large E-Commerce Retailer**”. I extend my appreciation to JD.com for making available the transaction level data that enabled me to conduct this project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through secondary sources and carefully analysed to assure its reliability.

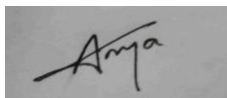
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all recommendations are business-specific and limited to this project exclusively and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:

A handwritten signature in black ink, appearing to read 'Arya', is shown within a rectangular box.

Name: Arya Sankhe

Date: 13/08/2026

1 Executive Summary

JD.com is one of the world's largest e-commerce companies, operating a first-party retail platform in China alongside a third-party marketplace, supported by its own nationwide delivery network.

The major business issues that the organization faces involve limited visibility across its delivery system, its discount use, and its paid membership programme. JD needs to understand why a large share of orders are served from distant warehouses rather than local ones, and what this pattern may cost the company in extra delivery time. It also needs to see which specific discount types do not drive higher sales volume, and whether paid members genuinely spend enough more than ordinary customers to justify what the programme costs to run.

The issues will be addressed by analysing a March 2018 transaction dataset of 5.5 lakh orders using descriptive analytics in Power BI and Google Sheets. The transaction data will be cleaned, joined, and evaluated using pivot tables, summary counts, and bar charts to measure delivery times, compare discounts, and benchmark member spending. The expected outcome helps the organization see where its delivery promise is weakest, which discounts are least effective, and how much its membership base is worth. This analysis enables JD.com to identify operational bottlenecks and provide clear data insights that would help it tackle a few of these issues.

2 Organisation Background

JD.com, also known as Jingdong, is one of China's largest retailers and among the biggest e-commerce companies in the world. It was founded by Liu Qiangdong in 1998 and moved online in 2004, growing from a single storefront into a nationwide platform. In recent years it has reported annual revenue well over 100 billion US dollars and serves several hundred million active customers.

What sets JD apart is how it sells. Under its first-party business it buys and owns its inventory outright, handling everything from pricing and stocking to delivery and after-sales service itself. Around this sits a third-party marketplace where independent merchants sell to the same customers. Its catalogue is wide, covering fresh food, apparel, electronics, cosmetics and everyday household goods.

Much of JD's reputation rests on logistics. It runs its own delivery operation through a large national network of warehouses and delivery stations, which is what allows it to offer the fast, same-day and next-day delivery it is known for.

3 Problem Statement, Background and Objectives

JD competes on delivery speed, discounts to drive sales, and runs a paid membership programme, but how well each of these is working is called into question. The reasons are partly internal and partly external.

Internally, the fulfilment network is large, and an order can be shipped from more than one warehouse. Cleaning the data made this concrete: nearly half of all orders (47.76%, about 2.6 lakh of 5.5 lakh) were served not from the warehouse nearest the customer but from a more distant one, which raises the question of what this pattern costs in delivery time. The discount data raised a second question. JD applies one discount type, the direct discount, far more than the others, which prompts the question of whether its most-used promotion is also its most effective at lifting how much customers buy. A third question concerns the paid membership programme: whether members, who pay for the service, actually spend enough more than ordinary customers to justify it.

Externally, customers expect fast delivery and JD's own promise reinforces that, competition makes every discount count, and holding on to higher-spending members matters in a crowded market. These early signals point to questions that only proper analysis can answer, which is what motivates the objectives that follow:

3.1 Fulfilment performance

To measure how reliably JD.com meets its promised delivery window, and to quantify the additional delivery time incurred when an order is served from a warehouse other than the one nearest the customer.

3.2 Promotion mechanism effectiveness

To compare the discount mechanisms JD.com applies across its first-party orders, and to identify which of them are associated with a higher average purchase quantity and which are not.

3.3 Membership customer value

To quantify the difference in average spend and units purchased between paid members and ordinary customers, and thereby establish what the membership programme is worth to the business.

4 Problem Solving Approach

4.1 Analytical methods and justification

The analysis is descriptive: it examines what the data shows rather than predicting or modelling. The raw tables are first cleaned by dropping records with missing timestamps, prices or key IDs, then joined on their order, product and customer IDs, and a few new columns are derived that the raw data does not hold directly like CrossRegion and Revenue.

To study fulfilment, each order's actual delivery time is worked out from its order and delivery timestamps and checked against the promised time to see how often the promise was kept; each order is also flagged as served from the nearest warehouse or a more distant one, so distant deliveries can be compared against local ones. To study promotions, first party orders are grouped by which discount type was applied, and average units bought are compared across groups. To study membership, average spend and units of paying members are compared against ordinary customers.

Each objective is shown mainly through bar charts and summary tables, for example the promise-met rate by order type, the average quantity bought by discount type, and average spend by membership status, with other chart types used where they present a comparison more clearly. Grouping and averaging, mostly through pivot tables.

4.2 Data source and rationale

The project uses secondary transaction level data released by JD.com for the 2020 MSOM Data Driven Research Challenge (Shen, Tang, Wu, Yuan and Zhou, 2020), accessed through a public repository mirror. It covers one anonymised consumable category over March 2018, with linked tables for orders, delivery, products and customers holding the timestamps, promised delivery time, warehouse fields, order type, quantity and discount breakdown the objectives needed. As it is a single baseline month with no major promotions, findings are a baseline snapshot, not generalised beyond the period or category present.

4.3 Analysis tools and justification

Because the raw data is large, Power BI with Power Query is used first to filter, join, derive fields and pre aggregate it into compact summary tables, since it handles data a spreadsheet cannot open directly. The summaries are then analysed and visualised in Google Sheets using pivot tables and charts. No programming or machine learning tools are needed, as the questions call for grouping and comparison, not modelling.

5 Expected Timeline

5.1 Work Breakdown Structure

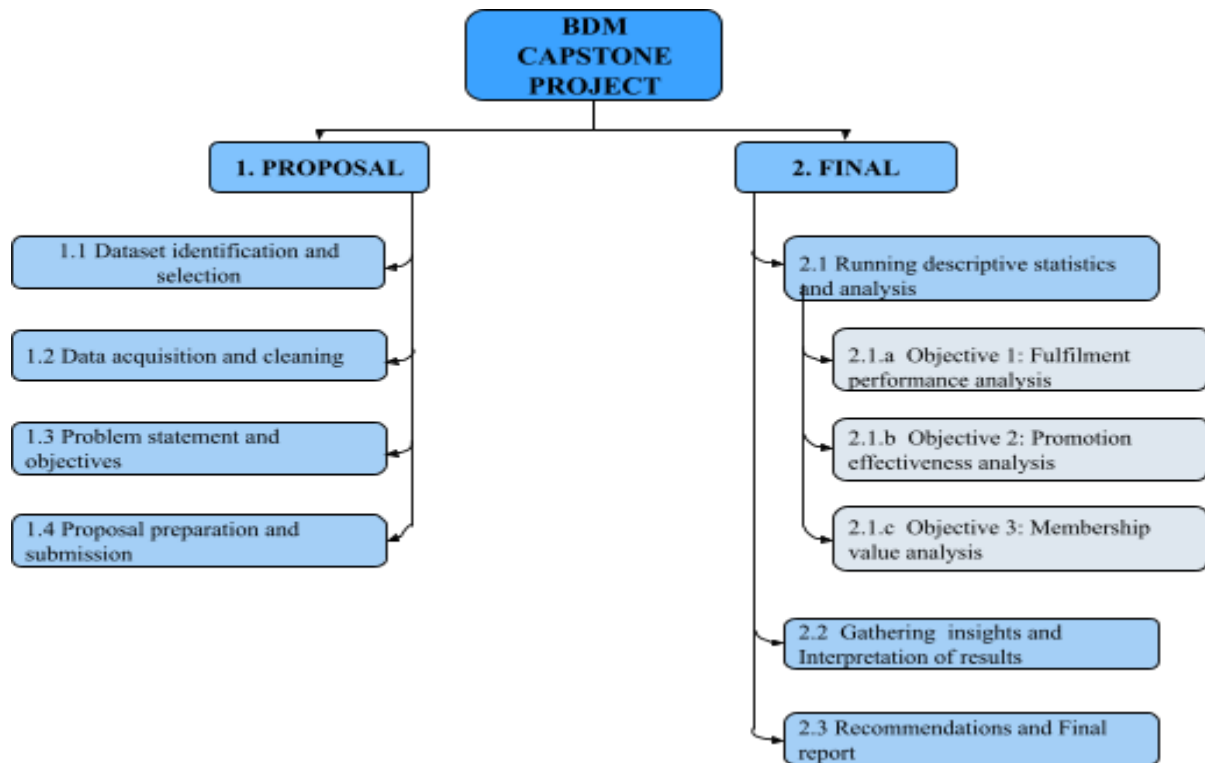


Figure 1: Work Breakdown Structure

5.2 Gantt Chart



Figure 2: Gantt Chart

6 Expected Outcome

6.1 Fulfilment performance

A clear view of how often JD keeps its delivery promise, and how much longer orders take when served from a distant warehouse, showing where the promise is most at risk and where JD might reconsider how stock is positioned.

6.2 Promotion mechanism effectiveness

An account of which discount types go together with customers buying more, and which do not, showing where JD's promotional effort is working and where it may be wasted.

6.3 Membership customer value

An estimate of how much more paying members spend than ordinary customers, showing what the membership programme is worth to the business.

Taken together, these findings will give the company a baseline understanding across three areas of its operation, so that it can protect its delivery standard, direct its promotions more effectively, and better appreciate its membership base. The project does not claim to resolve these issues in full; it aims to identify the bottlenecks and provide observations that would help the company tackle a few of them. All insights will be derived from the data and presented with the relevant figures and tables.